Does aging microplastics alter how particles affect the physiology of a freshwater copepod

The Zooplankton Ecology & Adaptation Lab At UMass Lowell

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Introduction

- Microplastics are increasing in abundance in aquatic systems
- Changes in food quality and quantity
- Copepods are important, highly selective aquatic grazers
- Biofilms can form on surface of microplastics
  - Nutritional benefit?
  - Increased exposure via reduced selectivity?

Central Question

Does microplastic aging alter how these particles affect the physiology of a common freshwater copepod?

Hypotheses

Hypothesis 1: Exposure to microplastics will decrease the thermal limits in the freshwater copepod *Skistodiaptomus pallidus.*

Hypothesis 2: The upper thermal limit for copepods exposed to aged microplastics will be higher than the upper thermal limit for those exposed to un-aged microplastics.

Hypothesis 3: Because of the nutritional content of biofilms, starved copepods with aged microplastics will exhibit a smaller decrease in thermal limit relative to the control copepods, while starved copepods with un-aged microplastics will exhibit a sharp decrease in thermal limits.

Preliminary Results

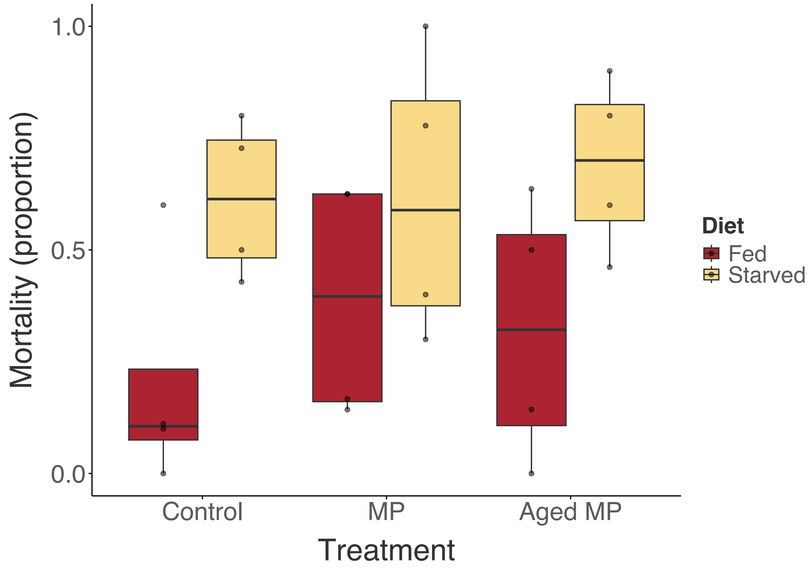


Figure 1: Proportional mortality in the different treatments. Survival was measured after one week. Each point shows mortality in one of the replicates, but replicates across experiments are pooled together. Mortality was consistently higher in the starved treatments.

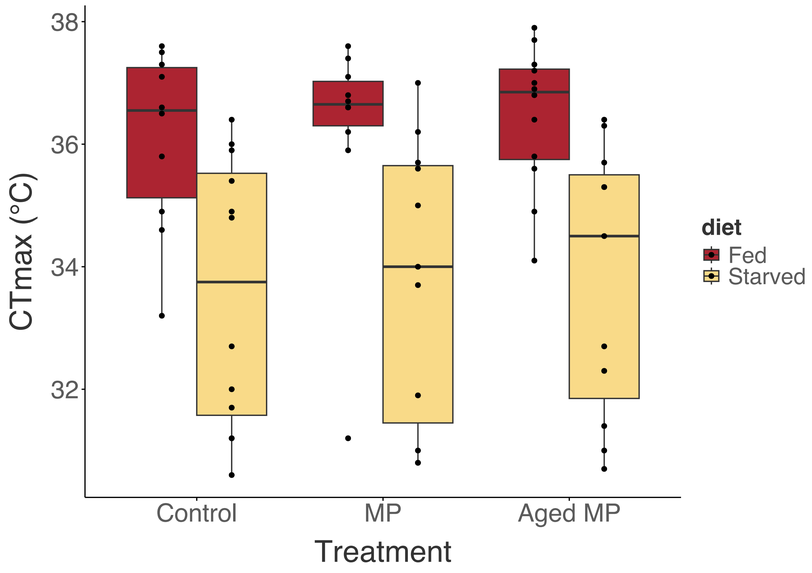
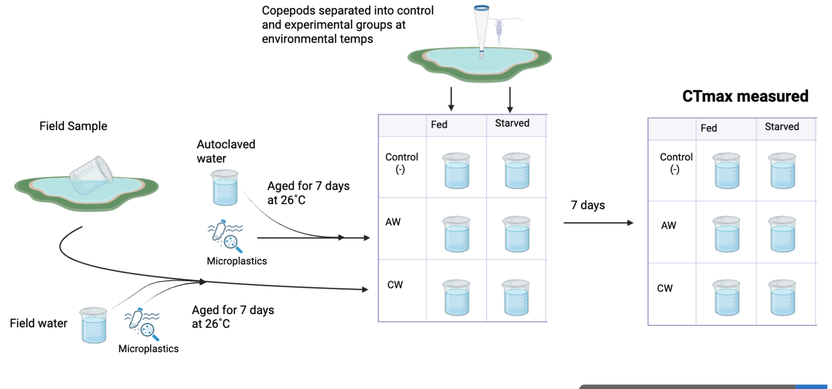


Figure 2: Upper thermal limits for copepods in the different treatments. CTmax was measured after one week. Each individual point represents the observed upper thermal limit of an individual copepod. Data from the two experimental replicates is pooled.

Methods



Conclusion

- Starvation has major effects on both survival and thermal limits
- Realistic microplastic concentrations have minimal effects on thermal limits, but may influence survival
- Survival seems to be highly variable from week to week

Next Steps

- Additional experimental replicates
- Dose-dependence of effects

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